Publication-Readiness Audit

Exact Query Bounds in Characteristic-Two Modal Computation (P02)

Independent analytical audit by ChatGPT

27 September 2026

Executive verdict

Recommendation: retain the publication project and revise the existing paper, rather than restart it. The core mathematics is considerably stronger than the standalone evidence package.

The manuscript has a viable center: an exact n-query lower bound for recovery of an n-bit Bernstein–Vazirani secret in the stated characteristic-two linear-modal model, including arbitrary finite ancillas and finite adaptive complete instruments. Standard XOR access attains that bound. I reconstructed the argument, examined its treatment of classical histories and vanishing branches, and obtained an alternative proof using the degree of a single secret-indicator polynomial. I did not identify a fatal error in any of the seven numbered mathematical results.

This is not an unconditional approval of the archive for submission. The original experimental supplement, canonical runner, historical datasets, and several referenced provenance records are absent. The manuscript also retains an author-placeholder footnote and internal publication-workflow prose. Its theorem-level priority remains unresolved: several ingredients are definitely prior art, while a source with the full claimed adaptive BV or generalized DJ statement was not located in the primary-source passages inspected. That negative search result does not certify novelty.

Weighted readiness: 7.388/10, reported as 7.4/10. This is an editorial assessment of the submitted archive, not a probability of acceptance, not a proof certificate, and not a score for a hypothetical repaired paper.

The strongest result is Theorem 4.2, supported by Lemma 4.1. The Deutsch and Deutsch–Jozsa arguments are sound and compact. The ring proposition is also correct, but is secondary to the paper's query-complexity contribution. The weakest parts are the unsupported portability of historical experimental claims and the unresolved distinction between a useful new quantified theorem and a straightforward specialization of known methods.

The best present form is a **focused technical note**. Put exact BV recovery and the adaptive recording argument at its center; retain Deutsch/DJ as clean related obstructions; keep UNIQUE-SAT as a short, explicitly attributed positive control. The ring example can be a compact appendix. The small Simon experiment belongs in the reproducibility supplement unless it carries a specific explanatory burden.

Decisions

Gate	Decision	Principal reason
A: mathematical integrity	**PASS WITH MINOR REPAIRS**	Numbered results survive reconstruction; clarify primitive eigenvector terminology and a few model conventions.
B: novelty/priority	**MATERIAL PRIOR-ART UNCERTAINTY**	Known foundations and methods are identifiable; exact theorem-level priority is not cleared.
C: evidence/reproducibility	**MATERIAL GAPS**	Build and integrity checks pass, but original experiment sources and outputs are missing.
D: preprint	**READY AFTER MINOR REPAIRS**	A proof-centered release is feasible after metadata, terminology, and truthful supplement repairs.
E: peer-reviewed submission	**READY AFTER TARGETED REVISION**	Needs explicit contribution comparisons, portable evidence or removal of unsupported empirical claims, and editorial cleanup.

Gate D is conditional on the minimal release route described below: do not claim an included or accessible supplement that has not actually been supplied. Gate E is not permission to ignore Gate B. A matched prior theorem could require substantial repositioning and a new significance assessment.

What this audit did, and did not, establish

The archive was recursively inventoried. The current source and all substantive supporting reports were inspected; historical and later review records were compared rather than accepted as verdicts. The manuscript was independently rebuilt to nine pages. Its extracted word sequence matches the original after whitespace normalization. Fresh standard-library verification code reproduced several reported finite results and exercised additional cases.

The audit is the work of one assistant applying several adversarial perspectives. It is not eight independent reviewers, human peer review, exhaustive literature clearance, or machine-checked formal verification. The supplied manuscript was not edited. Newly derived mathematics is isolated in the research-extension section and is not counted as a contribution of the submitted draft.

Deliverables and limitations

The report, proposed edits, ledgers, scorecard, and decisions are in report/; fresh checks and results in checks/; source inventory, hashes, and build comparisons in evidence/; unmodified manuscript sources and compilation logs in build/. The README supplies commands. Historical evidence and theorem-level priority remain unresolved. The optional rank proposition is separate audit work, not a claim attributed to the submitted paper.

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Scope, inventory, and source hierarchy

The evaluated upload is ExactQueryBoundsinCharacteristicTwo(P02)(1).zip. The canonical manuscript is the main.tex inside 01_SOURCE_PACKAGE_FROM_PORTFOLIO_20260921/source_package/; its title and PDF match the requested paper. It is dated 19 September 2026. A later viability assessment is dated 26 September 2026. Neither the folder date nor the date of that later assessment makes the manuscript a new revision.

For compact references below, **M:Lx–Ly** denotes the corresponding original main.tex lines, preserved in evidence/main_source_numbered.txt. Supporting files are identified by filename and heading. External primary sources are identified by S1–S11 in the source register. Fresh results are in checks/fresh_results.json. These are three distinct evidence classes.

Archive coverage

The ZIP has 40 entries: 33 files and seven directories, with 508,487 uncompressed bytes. It contains 14 Markdown files, six CSV files, five JSON files, four text logs, two TeX files, one bibliography file, and one PDF. There is no executable experiment source, historical experiment dataset, or nested ZIP.

The complete filename, size, and SHA-256 inventory is supplied as evidence/archive_inventory.json. The following groups were inspected:

Group	Contents and audit treatment
Canonical paper	main.tex, main.pdf, references.bib, phase_table.tex; statements, proofs, references, and rendered presentation.
Claim tracking	FINAL_CLAIMS.md, CHANGED_CLAIMS.md, theorem/proof/source crosswalk, novelty ledger; compared with actual statements.
Prior review	ADVERSARIAL_REVIEW.md, GATE_4_PRIORITY_PROOF_REPORT.md, unresolved-issues register; tested against the current source.
Literature trail	Primary-source review and search-log CSVs; treated as leads, not fresh evidence.
Build/provenance	Build manifest, four build logs, table provenance, visual-QA JSON, reproducibility readme; checked for present and missing dependencies.
Packaging/history	Top-level readmes, uniqueness summary, source manifest, historical context and manifest, review/reproducibility folder readmes.
Later assessment	RESEARCH_VIABILITY_REVIEW_2026-09-26/ assessment, decision, and manifest; evaluated independently.

The current readmes identify one canonical source package. Historical context is not an alternative manuscript. The later review adds opinions and references to spot checks, but does not supply a replacement manuscript or its external spot-check implementation.

Integrity and build

The root source manifest has 29 entries; all 29 file sizes and hashes match. Its coverage excludes the manifest itself and three later viability-assessment files. The historical manifest's 23 entries match after accounting for the documented relocation of the historical context file. The two files covered by the viability manifest also match. The original build manifest's manuscript-source and PDF hashes agree with the archive.

The manuscript rebuilt successfully using pdfLaTeX, BibTeX, and two further pdfLaTeX passes. The local bibtex symlink was broken; invoking the installed bibtex.original executable resolved that environment issue. The final log has no unresolved citations, undefined references, or overfull/underfull-box warnings. Both PDFs have nine pages and the same whitespace-normalized extracted word sequence. Binary PDF equality is neither obtained nor expected across TeX installations. Evidence is in build_verification.json and the build logs.

These checks establish file integrity and buildability, not mathematical correctness or authenticity of historical experimental runs.

Missing dependencies

REPRODUCIBILITY.md, TABLE_PROVENANCE.json, and associated records point outside this standalone archive. Missing items include the portfolio-level reproduction envelope and supplement, the canonical runner and 21-stage logs, the table-generation script, the oracle-contract JSON, small-algorithm and phase JSON files, UNIQUE-SAT CSV, Simon JSON, description-readout JSON, and the later spot_check_results.json. Some references use a sibling portfolio directory or an author's Windows path.

Consequently, M:L208's claim that commands, pinned environment, hashes, and stage times are in the supplement cannot be verified from the supplied package. The correct finding is **missing evidence**, not fabricated evidence or a false computation. The precise historical count of 552 degenerate oracle matrices remains unverified because its enumeration domain is not reconstructible from the delivered files.

Publication-readiness scorecard

The supplied prompt's weights are used unchanged. Scores are judgments with stated confidence, not measurements. The reproducibility score assesses the archive as delivered; the fresh verifier is a repair resource, not retroactive proof that the absent historical run occurred.

Category	Weight	Score / 10	Confidence
Mathematical correctness	20%	9.0	High
Completeness of proofs/results	8%	8.8	High
Model and assumption discipline	8%	8.8	High
Novelty and priority	12%	6.0	Moderate
Scientific significance	8%	6.5	Moderate
Literature coverage and positioning	8%	7.0	Moderate
Reproducibility	7%	4.5	High
Internal consistency	6%	8.0	High
Resolution of prior reviewer concerns	5%	8.0	Moderate
Exposition and readability	5%	7.2	Moderate
Scope and structural focus	4%	7.0	Moderate
Evidence and citation quality	4%	6.0	High
Submission hygiene	3%	5.5	High
Venue/form fit	2%	7.0	Moderate

The largest deductions are for unresolved priority, the missing original experiment envelope, and submission metadata. Mathematical correctness is high because the proofs themselves withstand reconstruction. Category-specific justifications follow and are preserved in scorecard.json.

Category	Basis for score
Mathematical correctness	All seven numbered results survived proof reconstruction; no substantive counterexample found.
Completeness of proofs/results	Proofs cover the stated quantifiers; the DJ optimum is explicitly not claimed.
Model and assumption discipline	Fields, complete histories, and lower/upper oracle interfaces are separated; minor terminology repair remains.
Novelty and priority	Closest ingredients are known; exact fully quantified BV/DJ priority remains unresolved after bounded search.
Scientific significance	A useful focused exact-query result, not yet a broad complexity theory or major algorithmic breakthrough.
Literature coverage and positioning	Main modal antecedents included; add precise instrument antecedent and nondeterministic-degree comparison.
Reproducibility	Manuscript builds and hashes match, but the original experimental runner and datasets are absent.
Internal consistency	Theorem scopes and ledgers largely agree; supplement/provenance wording exceeds the standalone delivery.

Category	Basis for score
Resolution of prior reviewer concerns	Substantive proof concerns are addressed; priority and portable evidence remain open.
Exposition and readability	Compact and readable, but needs a worked adaptive example and less internal-process language.
Scope and structural focus	BV is the natural center; ring/Simon material can move to appendices or supplement.
Evidence and citation quality	Primary sources support the main contrasts; historical finite evidence is not fully inspectable.
Submission hygiene	Placeholder author note, empty PDF author metadata, draft workflow prose, and bibliography cleanup remain.
Venue/form fit	A focused technical note is plausible; contribution scale and foundational motivation need sharper positioning.

$$\text{Weighted score} = \sum_i \frac{w_i s_i}{100} = 7.388 \quad \text{out of } 10.$$

No numerical average overrides an incorrect principal theorem, substantial anticipation by prior work, or a materially misleading evidence statement.

Actual contribution and claim ledger

The paper's demonstrated result is not that all modal computation is weak, all finite-field quantum algorithms fail, or all ring-valued phase information is useless. It establishes particular exact lower bounds in a specified linear access/readout model, with matching QXOR upper bounds only for Deutsch and BV.

The complete machine-readable ledger includes claim type, assumptions, status, novelty assessment, evidence, and action. The compact ledger below includes every numbered result and the major unnumbered mathematical, empirical, and interpretive assertions.

ID	Claim and source location	Audit result
C01	Characteristic-two fields, finite node spaces, complete histories; M:L34–48	Clear operative assumptions; finite field not required.
C02	Lower-bound class is broader than QXOR upper-bound interface; M:L44–55	Correct and important.
C03	Equal actions provide no information; M:L55	Correct for all query counts. Historical 552 count is separate.
C04	Exact discrimination, Lemma 2.1; M:L66–71	Verified; genuine direct sum, not pairwise disjointness.
C05	Complete recorded evolution, Lemma 2.2; M:L73–78	Verified, including secret-dependent zero branches.

ID	Claim and source location	Audit result
C06	Deutsch one-query impossibility, Theorem 3.1	Verified over the stated broad pointwise class.
C07	Deutsch exact QXOR complexity two	Verified lower and elementary matching upper bound.
C08	DJ one-query impossibility, Theorem 3.2	Verified for $n \geq 2$; $n = 1$ is Deutsch.
C09	DJ upper bound $2^{n-1} + 1$ under QXOR	Verified; not a modal optimality theorem.
C10	Recorded monomial lemma, Lemma 4.1	Verified, with at most $\sum_{j \leq q} \binom{n}{j}$ coefficient vectors.
C11	BV lower bound $q \geq n$, Theorem 4.2	Verified; strongest core result.
C12	BV QXOR upper bound n	Verified by basis-address queries.
C13	Inverse and node-specific fixed actions retain the lower bounds	Verified by the same affine dependence and branch induction.
C14	Ring kickback $X\chi = z\chi$; Proposition 5.1	Verified in the explicitly separate A-module example.
C15	Analyzer diagonal form and binary rank $4 + 2n$	Verified analytically and for six fresh finite cases.
C16	Order-two character obstruction; M:L183	Verified; does not cover arbitrary groups/transforms.
C17	Primitive-vector obstruction for eigenvalue R; M:L194	Correct intended statement; terminology needs repair.
C18	Four-cycle primitive R-eigenvector and changed oracle	Verified; not the same query interface as QXOR.
C19	Compute/control/uncompute costs two QXOR calls	Correct for that construction; not a proven universal minimum.
C20	Tensor-base distinction; M:L196	Verified; balancing over A differs from binary tensoring.
C21	Known one-query UNIQUE-SAT and displayed formula	Verified and correctly attributed, not novel.
C22	Deutsch 15/255 preparation counts	Freshly reproduced for standard QXOR.
C23	Historical 552 degenerate matrices	Unverified historical enumeration; source dataset absent.
C24	Historical small BV identities	Unverified exact historical scope; fresh independent tests are available.
C25	UNIQUE-SAT 132 instances	Freshly reproduced for $n = 1, \ldots, 6$.
C26	Simon 65,535-preparation, 36-oracle scan	Freshly reproduced at exactly the stated narrow scope.
C27	Six phase ranks and 192 primitive-vector candidates	Freshly reproduced.
C28	Full-description readout is extra access	Correct conceptual distinction; historical ablation instances absent.
C29	Historical full canonical suite/provenance	Not reproducible from this ZIP; repair claim or delivery.

ID	Claim and source location	Audit result
C30	Prior qualitative/methodological antecedents	Supported by primary comparisons; exact priority still unresolved.
C31	No dependence on companion CM/chain-ring theorems	Confirmed for the numbered proofs.
C32	Current format and publication status	A technical note is a reasonable judgment, not a proved scholarly entitlement.

The main claim files and crosswalk largely agree with the manuscript. The serious discrepancy is not an inflated DJ theorem or a hidden QXOR assumption in the lower bound; it is the difference between the archive's implied reproducibility envelope and what the recipient actually receives. A second issue is the ambiguous phrase "primitive eigenvalue," discussed below.

Theorem-by-theorem proof audit

Exact discrimination: Lemma 2.1

Status: verified. Let $U_i = \operatorname{span} S_i$. A complete readout, with all outcomes grouped by label, gives an injective linear map $L : V \to \bigoplus_i W_i$. Exactness gives $L(U_i) \subseteq W_i$. If $\sum_i u_i = 0$, then $\sum_i L(u_i) = 0$ in disjoint coordinate groups, so every $L(u_i) = 0$ and injectivity gives every $u_i = 0$. This proves directness, including for more than two labels.

Conversely, concatenate bases of the U_i and extend to a basis of V. The complete dual-coordinate instrument is injective. Each required state has support only in coordinates assigned its own label; coordinates in the complement vanish on all required states and can receive any label. Multiple states per label cause no problem. An explicit finite-family convention would improve readability; with finite-dimensional V, a direct sum of nonzero summands is necessarily finite anyway.

The familiar singleton criterion is prior art (S4, Section 4). The set-valued statement and its proof are elementary linear algebra, not a persuasive independent novelty centerpiece.

Complete recorded evolution: Lemma 2.2

Status: verified. At a leaf, take the identity. If the descendant record maps T_b are injective and the instrument stack $(L_b)_b$ is injective, then

$$(T_b L_b v)_b = 0 \Longrightarrow L_b v = 0 \text{ for all } b \Longrightarrow v = 0.$$

Composition with an injective gate preserves injectivity. For each fixed oracle its query maps are invertible, so the same induction shows that the overall record of a nonzero initial state cannot vanish. It is not asserted that one oracle-independent evolution map describes all secrets.

Different terminal dimensions are harmless because the common bookkeeping space is their direct sum. A zero branch remains a zero block. Early stopping produces a leaf with the identity as its continuation. Fixed ancilla addition is an injection over a field. Finite trees allow the induction to terminate. This is not a theorem about an unbounded stopping process.

The record is a mathematical representation of all possible histories; the proof does not require a physical operation that coherently recombines classical outcomes. No hidden interference between distinct histories is used. The model's completeness condition is equivalent to the common-kernel condition in the established generalized-modal framework (S2, equation (47)); cite that exact antecedent at the definition.

Deutsch: Theorem 3.1

Status: verified. Write a nonzero preparation as (a_0, a_1) in the two address sectors and let $\Delta = G_0 + G_1$. The four postquery states satisfy

$$v_{00} + v_{11} = v_{01} + v_{10} = (\Delta a_0, \Delta a_1) = w.$$

If $w \neq 0$, the constant and balanced spans overlap nontrivially. If $w = 0$, the separate address blocks imply $\Delta a_0 = \Delta a_1 = 0$, so all four states coincide. They remain nonzero because the query is invertible. Lemma 2.1 rules out exact classification in either case; Lemma 2.2 prevents query-free complete postprocessing from removing the obstruction.

Before the first query, branch states are secret-independent. A possible query-free terminal branch cannot be correct on both promise classes. Thus any exact one-query algorithm would have a nonzero prequery branch, possible for every oracle, to which the argument applies. This addresses adaptivity without silently replacing it by a nonadaptive model.

Arbitrary finite ancillas are already included in the a_x. Commutation of G_0, G_1 is not used. Inverse queries simply substitute another fixed invertible pair. Preparation on one address sector, one-dimensional targets when possible, and equal oracle actions all remain covered.

The two-query QXOR upper bound uses basis inputs at the two addresses, retains their Boolean answers, and computes their parity. It does not require cloning an unknown modal state. The qualitative antecedent is explicit in James–Ortiz–Sabry (S3, conclusion); the present all-ancilla adaptive statement must be compared separately rather than advertised as the first discovery that familiar Deutsch interference fails.

Deutsch–Jozsa: Theorem 3.2

Status: verified as a one-query obstruction. For $n \geq 2$, partition the $N = 2^n$ addresses into equal blocks A, B, C, D. The indicators of $A \cup B$, $A \cup C$, and $B \cup C$ are balanced and have Boolean XOR zero. In characteristic two, pointwise evaluation gives

$$O_{A\cup B} + O_{A\cup C} + O_{B\cup C} = O_0.$$

At an address in one of the first three blocks, two G_1 contributions cancel and one G_0 remains; at an address in D, three G_0 contributions give G_0. The identity is independent of target dimension and preparation. Applying it to a nonzero preparation puts its nonzero constant-zero output in the span of balanced outputs. Exact discrimination is impossible. The same secret-independent prefix reasoning handles one-query adaptive trees.

Distinct deterministic QXOR queries at $N/2 + 1$ addresses give the stated upper bound: a balanced function cannot agree at all those addresses. The manuscript correctly concludes only

$$2 \le Q_{\text{exact}}^{\text{QXOR}}(\text{DJ}_n) \le 2^{n-1} + 1.$$

The gap is an open task within this paper, not a proof gap in a claimed optimum. No finite experiment in the package closes it. Positive odd-characteristic constructions in S6 and S7 use different scalar assumptions and are not counterexamples.

Recorded monomials: Lemma 4.1

Status: verified. For a BV secret $s \in \{0,1\}^n$ and $f_s(x) = s \cdot x$, Boolean parity embeds as ordinary field addition specifically because the characteristic is two. Thus each oracle node has

$$O_s = O_0 + \sum_{j=1}^{n} s_j A_j, \qquad A_j = \bigoplus_x x_j (G_0 + G_1).$$

Fix a path through the syntactic tree. At that path the matrices are fixed, even though another history would select different ones. The initial coordinates have degree zero in s. A secret-independent linear transformation does not increase degree; an oracle increases it by at most one. Reducing $s_j^2 = s_j$ is an identity of functions on the Boolean secret cube, not an algebraic relation imposed on every element of F.

For a leaf h with at most q calls, write its vector as a sum of squarefree monomials with fixed coefficient vectors. Use one finite common direct sum over all syntactic leaves and concatenate the coefficient vectors for each monomial across those leaves. Impossible histories contribute zero values; they are not deleted depending on the secret. Early leaves have lower degree, which is permitted. Therefore

$$\Psi_s = \sum_{\substack{J \subseteq [n] \\ |J| \le q}} s_J c_J, \qquad \dim \operatorname{span}\{\Psi_s\} \le D(n,q) := \sum_{j=0}^{\min(q,n)} \binom{n}{j}.$$

The number $D(n,q)$ is the dimension of the available Boolean polynomial function space. It is an upper bound on this particular algorithm's record span, not a guarantee that its coefficient vectors are independent.

Branch-dependent pairs and inverse access retain affine dependence. Finite ancillas change coefficient-vector dimension but not the number of monomials. Recorded histories do not supply uncharged secret bits: their possibility patterns are the supports of the same polynomial vectors. Histories that reconverge in an implementation may be unfolded into a finite tree; forgetting them cannot create an exact algorithm unavailable with them retained.

The underlying amplitude-degree induction is an established method (S5, Lemma 4.1, whose proof explicitly notes that linearity rather than unitarity is used). The characteristic-two parameter substitution and the complete-history formulation are the paper's relevant specialization, not a wholly new polynomial method.

Bernstein–Vazirani: Theorem 4.2

Status: verified. For each secret s, completeness makes $\Psi_s \neq 0$. Exactness makes it supported only in terminal blocks carrying label s. Hence the 2^n records are linearly independent. They are also contained in the span of at most $D(n,q)$ coefficient vectors. When $q < n$, $D(n,q) < 2^n$, a contradiction.

This is stronger than showing that one familiar Fourier circuit fails. It rules out every algorithm in the stated finite adaptive contract. It does not extend to nonlinear secret-dependent gates, postselection, free description access, or a changed oracle primitive. In particular, over characteristic zero the Boolean parity $s \cdot x$ is not generally a degree-one scalar polynomial in the bits s_j; the proof does not contradict ordinary quantum BV.

The upper bound is exactly n QXOR queries at the unit addresses e_j, followed by basis-coordinate readout. General pairs are only in the lower-bound class. Some such pairs encode no information and have infinite complexity for recovery. The manuscript makes that distinction correctly.

Independent alternative proof

There is a useful shorter audit proof. Fix one secret s^0. Some terminal coordinate $p(s)$ in a leaf labelled s^0 has $p(s^0) = a \neq 0$. Exactness gives $p(s) = 0$ for all other secrets, and the branch-degree argument gives $\deg p \leq q$ after multilinear reduction. Uniqueness of Boolean multilinear interpolation gives

$$p(s) = a \prod_{j:s_j^0=1} s_j \prod_{j:s_j^0=0} (1 + s_j).$$

Its degree is n, since the coefficient of $s_1 \cdots s_n$ is $a \neq 0$. Therefore $q \geq n$.

This proof is independently derived here, not a claim of historical novelty. It is a useful cross-check because it does not use a global dimension comparison or a physically coherent interpretation of the record. A brief comparison with the nondeterministic-polynomial literature is appropriate (S10); its ordinary complex, positive-acceptance model is not the same as all-branches-correct modal recovery.

Ring example: Proposition 5.1 and adjacent claims

Status: proposition verified; adjacent primitive terminology needs clarification. In $A = \mathbb{F}_2[u]/(u^4)$, $z = 1 + u^2$ has $z^2 = 1$, so $X(1,z)^T = (z,1)^T = z(1,z)^T$. Elementary invertible row and column additions give

$$H_z = \begin{pmatrix} 1 & 1 \\ 1 & z \end{pmatrix} \sim \operatorname{diag}(1, u^2).$$

Tensoring over A gives diagonal factors u^{2k} indexed by subsets of size k. All terms for $k \geq 2$ vanish. Multiplication by 1 has binary rank four, and multiplication by u^2 has binary rank two. There is one former term and n latter terms, proving binary rank $4 + 2n$ in dimension $4 \cdot 2^n$.

The group-character argument is correct: an element of a characteristic-two field satisfying $a^2 = 1$ must be 1. Over this particular local ring every unit reduces to 1 in the residue field, so a unit-valued

character matrix of size greater than one reduces to the all-ones matrix and is not invertible. This is not a prohibition on every finite-field transform or on odd-order character groups.

The phrase "R cannot be a primitive eigenvalue" at M:L194 should be replaced by a statement about a **primitive eigenvector**. With $R = 1 + u$, the equation $X\eta = R\eta$ forces $u^2\eta = 0$, excluding a vector with a unit coordinate. However, nonzero nonprimitive eigenvectors do exist. For example,

$$\eta = (u^2, u^2 + u^3)^T, \qquad X\eta = R\eta.$$

The fresh exhaustive check finds all three nonzero such vectors in A^2, and none among the 192 primitive vectors. This does not refute the intended proof; it shows why the qualifier must attach to the vector.

The four-cycle eigenvector has coordinates R^{-j} for $j = 0, 1, 2, 3$, all units, and the shift gives eigenvalue R. Its controlled oracle is a different primitive. The described compute/control/uncompute simulation consumes two QXOR calls; the paper does not prove that every possible implementation needs two. Keep that distinction explicit.

Finally, $\otimes_A$ allows moving an A-scalar between factors, whereas a literal independent binary tensor product does not. The warning is correct. The section expressly does not export the field-query theorems to arbitrary ring-state models. No field-style dimension argument over a ring is required for Proposition 5.1.

Contract matrix and boundary checks

Feature	Field lower bounds	Matching upper bounds	Ring example
Scalars	Any characteristic-two field	Same field, Boolean basis labels	$A = \mathbb{F}_2[u]/(u^4)$
States/spaces	Nonzero vectors; finite dimension per node	Basis-state constructions suffice	Free modules; specify primitive vectors where used
Composition	Tensor over F	Tensor over F	Tensor over A
Gates	Secret-independent invertible maps/injections	Fixed Boolean basis permutations and allowed readout	Invertible A-linear maps
Ancillas	Arbitrary finite	Finite work/answer storage	No universal ancilla lower bound asserted
Measurement	Finite stack-injective linear instruments	Complete basis readout	Nonzero contractions in worked example
Adaptivity	Finite recorded tree	Not needed for displayed optima	No general adaptive query theorem
Oracle	Fixed pointwise invertible actions; node-specific pairs allowed	Standard QXOR specifically	Flip or separately declared cyclic action
Inverses	Lower-bound argument persists	QXOR is its own inverse	Construction-specific

Feature	Field lower bounds	Matching upper bounds	Ring example
Exactness	Every possible output correct; no vanished total record	Same requirement	Analyzer claim is algebraic
Exclusions	Postselection, nonlinear updates, free coefficient table, unbounded stopping	Same exclusions	No silent scalar/tensor reinterpretation

A complete instrument is not a choice to keep only a successful outcome. Its stack has trivial kernel. This is decisive: allowing arbitrary postselection could invalidate the nonzero-record step. Arbitrary finite workspace is permitted, but no uniform gate-efficiency theorem is claimed. An exponential-size secret-independent measurement may be allowed by a pure query model; it would require separate treatment in a computational-complexity model.

The equal-action case $G_0 = G_1$ is a legitimate member of the lower-bound class, not evidence that QXOR has infinite complexity. Similarly, an inverse action remains affine in the Boolean control, but an entirely different phase/cyclic access primitive must not inherit the QXOR cost label without an implementation argument.

Counterexample and edge-case attack report

The following attacks were actually performed, either analytically or by the supplied fresh code. No exhaustive all-dimensional search is claimed.

Attack	Method and outcome
Pairwise versus genuine direct sum	Analytic reconstruction uses a full linear relation; no pairwise-only shortcut. Three binary lines can be pairwise disjoint without forming a direct sum, so the stronger condition is necessary.
Vanishing histories	Proved stack injectivity by backward induction; randomly tested finite trees with zero branches and early stops.
History as extra information	Fixed syntactic direct-sum record includes every leaf; secret-dependent support is captured by polynomial coordinates.
Arbitrary ancillary dimension	Analytic proofs use unrestricted finite sector vectors. Fresh dimension-eight Deutsch checks are only illustrations.
Equal/noncommuting actions	All 36 ordered pairs in $\mathrm{GL}(2,2)$ tested; proofs do not assume commutation or distinctness.
Larger characteristic-two field	2,000 seeded cases in $\mathbb{F}_4$ passed; general field coverage is analytical, not inferred from this sample.
One-dimensional target or sparse preparation	Deutsch overlap proof remains valid; no special support assumption enters.
Inverse queries	Replacing each action by its inverse preserves the same Boolean affine form.

Attack	Method and outcome
Early termination and node-specific pairs	Path-degree induction and random tree tests passed; syntactic padding introduces no new monomials.
$n = 1$ and $n = 2$	BV proof includes $n = 1$; DJ $n = 1$ reduces to Deutsch; four-block DJ relation tested at $n = 2$.
Infinite characteristic-two field	Proofs use finite-dimensional linear algebra and Boolean interpolation, not counting all field elements.
Ring nonprimitive vectors	Explicit nonprimitive R-eigenvector found; intended primitive-vector obstruction survives.
Shared versus literal tensor base	Analytic balancing check; binary rank calculation uses shared A multiplication, not independent nilpotents.
Simon's three output labels	Fresh scan tests genuine directness by comparing combined and separate span ranks, not pairwise intersections alone.

The source does not define $n = 0$ as a target task. It is harmless to state $n \geq 1$ explicitly for BV and Deutsch–Jozsa inputs rather than leave degenerate conventions implicit. No claimed bound depends on a nonstandard zero-bit convention.

Independent computations and reproducibility

Fresh verification scope

The fresh program checks/independent_checks.py uses Python's standard library, deterministic enumeration where stated, and recorded seeds where sampling is used. All eight top-level checks passed in this environment. Run it with:

```
python checks/independent_checks.py --output checks/fresh_results.json
```

The output records Python/platform information, code SHA-256, case counts, and runtimes. --skip-simon provides an optional reduced run. The program is new audit code, not the missing canonical runner.

Fresh check	Coverage	Result
Deutsch, dimension four	36 ordered invertible pairs x 15 nonzero preparations = 540 cases	No exact one-query solution
Deutsch, dimension eight	Same pairs x 255 preparations, with a two-dimensional untouched ancilla = 9,180 cases	No exact one-query solution
DJ relation	$n = 2$, 36 pairs x 255 preparations = 9,180 checks	Required three-balanced relation holds
Boolean evaluation ranks	$n = 1, \ldots, 7$ and $q = 0, \ldots, n$: 35 checks	All match $D(n, q)$
Adaptive trees	168 seeded tree degree checks; 480 fixed-secret stacked-map rank checks	Degree and injectivity tests pass
$\mathbb{F}_4$ Deutsch	2,000 seeded examples; $\lvert\mathrm{GL}(2, 4)\rvert = 180$	No sampled counterexample

Fresh check	Coverage	Result
Ring ranks	$n = 1, \ldots, 6$	Ranks $6, 8, 10, 12, 14, 16$
Ring eigenvectors	All 192 primitive vectors in A^2	No primitive R-eigenvector; three nonprimitive nonzero examples
UNIQUE-SAT	132 promised instances over $n = 1, \ldots, 6$	All protocol outputs satisfy exactness
Simon	All 65,535 nonzero preparations in binary dimension 16; 36 standard oracles, three shifts	No exact one-query, no-ancilla preparation
New extension check	All 255 nonempty promises on three Boolean addresses; $q = 0, 1, 2, 3$	1,020 feature/graph-state rank comparisons agree

The random adaptive tests use complementary-projection instruments, early stopping, branch-specific gates, and fixed invertible oracle pairs. They are not exhaustive over all complete instruments. The general induction, not these tests, establishes the theorem's quantifiers.

Historical evidence versus fresh reproduction

The standard Deutsch 15/255 counts, UNIQUE-SAT 132 count, Simon scan, phase ranks, and primitive-vector search now have fresh matching results. That increases confidence in those assertions. It does not validate the provenance of the historical JSON or a 21-stage suite that was not delivered.

The 552-matrix historical count, exact small BV dimensions from missing JSON, and exact description-ablation instances remain unverified. Their conceptual claims are separate: $G_0 = G_1$ is information-free by inspection of the oracle formula; BV has a proof independent of small identity checks; reading an entire classical coefficient description changes the access model. These facts do not validate undocumented enumeration counts.

The manuscript build is independently reproducible with the supplied source. Table values in phase_table.tex are mathematically reconstructible, but its claimed historical generation by complete_p02.py from rerun JSON is not reproducible until those inputs and generator are supplied.

The smallest adequate repair is to add a self-contained verifier, outputs, one-command instructions, and truthful provenance for the finite results retained in the paper. Alternatively, remove historical empirical claims that do not serve the proof-centered note. A full portfolio archive is not necessary if the paper's modest checks can be reproduced locally.

Prior art, novelty, and citation audit

Search method and limits

Fresh searches used theorem-content phrases about modal BV, exact characteristic-two Deutsch/DJ, adaptive polynomial lower bounds, finite-field oracle identification, modal state discrimination, and nondeterministic polynomial degree. Relevant primary documents were then opened directly and inspected at the sections below. Citation links were followed among the modal foundations, finite-field computation, state-discrimination, and polynomial-method sources.

Several broad exact-phrase searches returned noisy or relaxed matches, including ordinary complex quantum algorithms and unrelated uses of "oracle." Such results were not counted as evidence

about modal query lower bounds. This was a bounded literature investigation, not a comprehensive citation-database sweep. No equivalent full adaptive BV theorem was found in the inspected passages, but the search is insufficient to certify absence throughout the literature.

The source register and search log are separate deliverables. Their purpose is to make both the positive source matches and the remaining gaps inspectable.

Closest antecedents and consequences

Modal foundations and complete instruments. S1 supplies the basic finite-field modal framework. More specifically, S2 equation (47) requires a trivial common kernel for an unconditional family of linear operations, and Section 4.4 relates operator-family and enlarged-system descriptions. The manuscript's injective stack is the same common-kernel condition, since $\ker(L_b)_b = \bigcap_b \ker L_b$. Cite this precise link at M:L35–40; do not portray completeness itself as new.

Qualitative Deutsch obstruction. S3's conclusion already identifies loss of the familiar Deutsch advantage in its finite-field setting. It does not, in the inspected passage, state the present arbitrary-pair, all-finite-ancilla, finite-adaptive theorem. The appropriate positioning is a quantified contract-level formulation of a known qualitative obstruction, with exact priority still open.

State discrimination. S4 Section 4 explicitly connects modal distinguishability with linear independence and studies multiple-copy distinguishability. This undercuts any novelty claim for the singleton linear-independence criterion, while leaving the paper free to state and prove the useful labelled-subspace formulation. S4 is also relevant to the new graph-state tensor construction below, without itself supplying its oracle-query rank characterization in the inspected section.

Polynomial method. S5 Lemma 4.1 gives degree-at-most-query-count amplitude polynomials; the following page explicitly says linearity, not unitarity, is used in that step. Its probability-polynomial step is different and must not import an inappropriate factor of two into this modal argument. The method is known; the paper's contribution must lie in the parameterization, exactness model, quantified statements, and useful consequences.

Additional antecedent worth adding. S10 Theorem 2.3 identifies nondeterministic quantum query complexity with nondeterministic polynomial degree. Its support-based proof perspective is relevant to the alternative BV proof and to promise-specific extensions. However, it is an ordinary complex, positive-acceptance model with a different oracle-variable parameterization. Its Fourier-based upper bound does not transfer automatically to characteristic two. This is a comparison to make, not proof that P02 is already contained in S10.

Odd-characteristic algorithms. S6's discrete-computing discussion and S7 Section 4.4 use complexified finite fields, including $\mathbb{F}_9$, and Hermitian/unitary structure. Positive DJ/BV statements there are not statements in the characteristic-two contract audited here. Preserve the domain distinction; do not describe those sources as contradicting the paper.

Uniform modular computation. S8 treats valid state spaces and transformations and uniform circuit complexity over algebraic scalars. This supports the manuscript's warning that state admissibility over rings and computational complexity are distinct from a fixed oracle promise's query count. It is not a substitute citation for the present exact bounds.

UNIQUE-SAT. S9 contains the known one-query protocol reproduced at M:L199–206. Its circuit and two promise cases match the manuscript's algebraic presentation. Keep the attribution and positive-control role. Do not score it as an original algorithm.

Claim-level novelty assessment

Claim	Best present classification	Confidence
Basic modal linear computation	Known framework	High
Common-kernel / complete instrument condition	Known condition with explicit history bookkeeping	High
Singleton discrimination criterion	Known result	High
Labelled-subspace direct-sum lemma	Elementary extension/reformulation	High
Finite recorded injectivity	Elementary proof device under established completeness	High
Qualitative Deutsch failure	Known qualitative fact	High
Broad exact Deutsch theorem	Quantified refinement; exact priority unresolved	Moderate
Generalized one-query DJ obstruction	Apparently distinct formulation in inspected sources; priority unresolved	Moderate
Degree-per-query architecture	Known technique specialized to this setting	High
Fully adaptive BV n lower bound	Strongest candidate contribution; no matched theorem located, not certified new	Moderate
QXOR matching upper bounds	Elementary constructions	High
Ring character degeneration	Elementary algebraic obstruction with antecedents	High
Specific $4 + 2n$ calculation	Useful worked specialization; priority not independently established	Moderate
UNIQUE-SAT protocol	Known, reproduced result	High
Finite Simon enumeration	Bounded empirical observation; not a general theorem	High on scope, low on significance

"Historical firstness is not claimed" is sensible but does not by itself establish a research contribution. Before journal submission, state what remains beyond the closest sources in terms of hypotheses, conclusion, and proof content. A clean known-result exposition may still be valuable, but it has a different submission case from a new theorem.

Bibliography and wording repairs

The bibliography has nine entries. The principal sources are relevant, but theorem/page locators should replace some broad historical references. At least one published-version improvement is confirmed: *Modal Quantum Theory* appeared in *Foundations of Physics* **42**, 918–925 (2012), DOI 10.1007/s10701-012-9650-z; the manuscript gives only the earlier arXiv entry. Update that entry without confusing it with *Almost quantum theory*. The Beals journal citation is already substantially complete. Published replacements for the other arXiv entries were not all established in this audit; do not invent them.

Add S10 as a nearby method comparison if the alternative proof or rank extension is included. Consider original Deutsch, DJ, and BV citations when explaining the named tasks' origins. No

broad historical-priority assertion should rely on a search snippet or a local novelty label.

Suggested contribution wording:

> We prove exact oracle-query bounds for characteristic-two linear-modal computation under an explicitly recorded, complete-instrument contract. The amplitude-degree method and basic modal discrimination criterion are established; the contribution here is their explicit application to all finite adaptive histories and finite ancillas for these promise tasks. We obtain exact QXOR complexity two for Deutsch and n for BV, and a one-query DJ obstruction. The precise relationship of these quantified statements to prior formulations is detailed below.

Use this as conservative positioning, not as a substitute for the promised detailed comparison.

Reconciliation of prior reviews

Existing reviews were assessed as collections of claims, not independent votes. The archive itself labels the adversarial perspectives as agent-based; there is no evidence that a human specialist panel reviewed the paper.

Prior issue	Was it valid?	Current status
Parity must be degree one over the actual scalar field	Yes	Resolved: characteristic two is explicit, including infinite fields.
Arbitrary postselection could fake success	Yes	Resolved in the contract by stack injectivity and all-possible-label exactness.
Adaptive histories and zero branches need proof	Yes	Resolved analytically by Lemmas 2.2 and 4.1; fresh tests corroborate only examples.
Arbitrary ancillas cannot be inferred from finite scans	Yes	Resolved in proofs; finite tables remain narrowly labelled.
General-pair lower bounds do not imply general-pair upper bounds	Yes	Resolved explicitly; $G_0 = G_1$ counterexample is included.
Qualitative Deutsch failure has antecedents	Yes	Acknowledged; precise quantified priority still not cleared.
Fully quantified BV/DJ priority needs checking	Yes	Still open at theorem level; this audit narrows but does not close it.
Ring kickback might be confused with an invertible analyzer	Yes	Mathematical distinction resolved; primitive-vector wording still needs editing.
Free description inspection is not operational readout	Yes	Conceptual distinction resolved; exact historical ablation data missing.
Simon scan must not become an all-size lower bound	Yes	Scope is correctly restricted; fresh exact-scope scan agrees.
552 should refer to matrices rather than pairs	Plausible correction	Wording is corrected, but the historical enumeration itself cannot be verified.

Prior issue	Was it valid?	Current status
Reproduction records should support tables	Yes	Not resolved for a standalone recipient; external supplement is absent.

New or sharper findings from this audit are: the exact connection to S2 equation (47); the nearby nondeterministic-degree literature; the nonprimitive R-eigenvector illustrating the terminology risk; the distinction between a successful portable build and a missing experiment envelope; and the constructive identification-rank result below. None should be represented as a finding already established by the supplied prior reviews.

Exposition, structure, and submission hygiene

The paper is short, mathematically coherent, and largely self-contained. The main theorem proofs do not require unpublished CM or chain-ring results. The abstract's oracle distinction and the task table's separate lower/upper columns are strengths.

The title is accurate but distributes attention among queries, adaptivity, and kickback. A tighter title such as **Exact Query Bounds for Characteristic-Two Modal Computation** is sufficient. A subtitle can be dropped. The introduction should identify BV recovery as the strongest theorem within its first substantive paragraph and explain what arbitrary complete histories add beyond a failed textbook circuit.

Add one concrete $n = 2$ illustration. With one query, every terminal record is of the form $c_\varnothing + s_1c_1 + s_2c_2$. Consequently,

$$\Psi_{00} + \Psi_{10} + \Psi_{01} + \Psi_{11} = 0.$$

Four nonzero records confined to four different output labels cannot satisfy that relation. This single example explains the coefficient-space obstruction and why extra histories do not evade it. A small adaptive tree may be described in prose; a new decorative diagram is unnecessary.

Use "primitive eigenvector with eigenvalue R" consistently and define primitive as having at least one unit coordinate in this local free module. Call 2^n the free A-module rank rather than an unqualified vector-space dimension, and call $4 \cdot 2^n$ the binary dimension. Proposition 5.1's meaning is clear, but table headings should enforce the distinction.

The author footnote is a release blocker: an inherited label with pending metadata is not final authorship. Replace it with the intended publication name and actual affiliation/contact information, or use deliberate valid anonymous-submission formatting when a venue requires it. Do not invent those details. Populate PDF author metadata. Remove the inherited-project sentence and change the final section from "Scope and publication status" to "Discussion and limitations"; the author's internal gatekeeping belongs in the README, not the scientific argument.

The nine-page original was rendered for inspection. The overview shows no obvious overlap or clipping. The boxed figures largely repeat prose. They are optional. Some section and theorem transitions occur near page bottoms, and references spill into a sparse ninth page. These are polish

issues, not mathematical defects. The explicit `\clearpage` before the final section contributes to the sparse layout; remove it unless needed for a venue format.

Retain the positive control in a short paragraph or appendix. Move historical finite scans, especially Simon, into a properly supplied supplement. The ring example can remain if its role is stated as a counterexample to the inference "kickback implies invertible analysis," rather than a second main research theme.

Significance, format, and venue fit

A result can be sound without being major, and new without being important. The paper earns its strongest interest from excluding an entire class of finite adaptive modal BV algorithms, not merely a particular Fourier construction. It makes oracle and readout assumptions inspectable and shows why a characteristic-two linearization changes the familiar BV situation. Those are reasonable grounds for a short theoretical note.

The principal significance risk is that a specialist may view the result as a short corollary of standard amplitude-degree and modal-discrimination facts. The response should not be more peripheral examples. It should be a crisp theorem, an explicit comparison to the strongest antecedent, and one useful generalization or explanatory example if it genuinely strengthens the contribution.

The DJ gap does not by itself make the note incomplete. The ring example does not turn it into a full theory of computation over finite chain rings. The finite Simon scan is not strong enough to carry a standalone research claim here. UNIQUE-SAT establishes a useful contrast, but its novelty belongs to its source.

Plausible venue types are a foundational mathematical-physics journal that accepts rigorous model comparisons, or a theoretical/unconventional-computation venue that values precise algebraic query models. *Foundations of Physics* is a plausible scope match, not an acceptance recommendation: its current aims include foundational aspects of quantum and information theory (S11), and the modal framework has appeared there (S1). A submission would still need to explain the conceptual significance beyond an algebraic exercise. A mainstream quantum-algorithms venue would need especially clear motivation and a stronger account of why this nonstandard model is relevant.

No acceptance probability or prediction is offered. Editorial fit and actual novelty must be assessed separately from correctness.

Research extensions: one proved opportunity and one unresolved direction

A constructive exact-identification rank characterization

Status: independently derived in this audit; not part of the submitted manuscript; priority unresolved. The prior assessment correctly warned that an arbitrary rank bound is not automatically an achievable algorithm. For the paper's unrestricted QXOR and complete-readout model, a particular rank condition can in fact be made sufficient by an explicit construction.

Let X be a finite nonempty address set and P a finite nonempty family of distinct Boolean functions on X. Form $E_q(P)$ with columns indexed by $f \in P$ and rows indexed by subsets $S \subseteq X$ of size at most q:

$$(E_q(P))_{S,f} = \prod_{x \in S} f(x).$$

The empty product is one. In the manuscript's characteristic-two field model, with unrestricted finite-dimensional secret-independent linear operations and complete readout,

$$Q_{\text{exact}}^{\text{QXOR}}(\text{Identify } P) = \min\{q \geq 0 : \operatorname{rank}_F E_q(P) = |P|\}.$$

This is a statement about **identifying the entire promised function**, not arbitrary many-to-one classification, and not polynomial gate complexity.

Necessity

Write the truth-table bits as $t_x = f(x)$. Each pointwise query is affine in those bits. The recorded-path induction therefore expresses every output coordinate as a polynomial of total degree at most q in the t_x, reduced to squarefree form on Boolean inputs. The matrix of records has a factorization through $E_q(P)$, so its rank is at most $\operatorname{rank} E_q(P)$. Exact identification requires $|P|$ independent nonzero records. This necessity also holds for the broader fixed-action lower-bound class.

Sufficiency

A single QXOR call on the fixed nonzero preparation $\sum_{x \in X} |x, 0\rangle$ produces

$$|\Gamma_f\rangle = \sum_{x \in X} |x, f(x)\rangle.$$

Preparing q independent registers and querying each once produces $|\Gamma_f\rangle^{\otimes q}$. Preparation is permitted: any desired nonzero vector can be the image of a fixed basis vector under an invertible linear map. The tensor state is nonzero over a field.

Let B_q have these tensor states as columns. Every coordinate of a column is a product of q factors, each either $f(x)$ or $1 + f(x)$; hence the row space of B_q is contained in the row space of $E_q(P)$.

Conversely, fix a row monomial $\prod_{x \in S} f(x)$ with $|S| = k \leq q$. Assign k query slots to those addresses and take target label 1 in each. In the remaining slots use any fixed address and sum the two target-label coordinate rows. Each unused slot contributes $(1 + f(x)) + f(x) = 1$. Thus the desired monomial is a linear combination of rows of B_q. For $q = 0$, both matrices reduce to the constant feature. Therefore their row spaces, and ranks, agree.

If $E_q(P)$ has full column rank, the tensor graph states are independent. Extend them to a basis and apply the complete dual-basis readout, assigning one label to each promised function. This instrument is fixed from the known promise family and does not depend on the unknown function. It identifies every promised input exactly. The linear combinations in the row-space argument are a proof of rank equality, not a permission to inspect unknown coefficients for free.

Consequences and cautions

The result supplies a nonadaptive optimal-query realization for identification in this permissive model. Arbitrary adaptive protocols cannot beat the rank threshold. For a promised subset of BV secrets, the address family contains the unit queries, so the degree-q truth-query features span exactly the restricted degree-q secret monomials. The same criterion therefore gives an exact promised-BV identification characterization, not only a necessary bound.

For the full BV cube, the threshold is n, recovering Theorem 4.2. The fresh program checks row-rank equality on all 255 nonempty promises on three Boolean addresses at four query budgets. Those computations are corroboration; the argument above is the proof.

This does **not** give matching upper bounds for arbitrary pairs G_0, G_1, efficient gate synthesis, a uniform algorithm from a succinct promise description, or a general decision/classification characterization. The necessary measurement may be very large. Nor does it close the DJ gap, because DJ identifies a class rather than the entire truth table.

This is the highest-value optional strengthening found in this audit: it directly generalizes the paper's main theorem instead of adding a new topic. Before insertion, audit its novelty against oracle identification, modal multiple-copy discrimination, and algebraic feature-rank literature. It has not received an external review. Keep it as an optional proposition until that comparison is done.

Deutsch–Jozsa beyond one query

The natural next lower-bound question is whether low-degree coordinate polynomials can vanish on the balanced slice while remaining nonzero on constant inputs, and whether compatible complete instruments can realize those supports. This separates an algebraic necessary condition from an operational construction. A polynomial that separates the promise classes is not automatically an exact modal algorithm with the same query count.

No improved general DJ bound is established by this audit. Do not add one to the paper based on the identification proposition or a small enumeration. Closing the gap could substantially strengthen the research, but it is not needed to make the current limited theorem correct or to release a focused note.

Other directions

Learning k prescribed independent linear forms of a BV secret has an elementary k-query upper bound and a k-bit restriction lower bound. For the latter, restrict the secret to a k-dimensional affine slice on which the requested forms are bijective; the same degree argument applies. This is a useful corollary, not a separate research program. A short access taxonomy may help exposition, but the existing contract table already does much of that work. Expanding into unrestricted ring teleportation, other CM papers, or general physical interpretations would weaken the present focus.

Ranked revision plan

Each item is actionable. "New mathematics" refers to required work, not rewriting an existing proof. The archive contains a separate revision register with all fields.

Stage 1: integrity and release repairs

R1 – High; M:L17–18 and PDF metadata. Replace inherited author-placeholder text with intended final authorship or deliberate anonymous formatting. Remove pending-venue language. No new mathematics, literature, or computation. Small effort, requires author-supplied facts.

R2 – High; M:L208–223, phase-table caption, reproducibility records. Supply a portable supplement containing the source and exact data behind retained finite claims, or replace/remove claims whose historical evidence is missing. Use the fresh verifier only with fresh provenance. New mathematics: no. Literature: no. Computation: yes if historical claims remain. Effort: small for a proof-centered release, moderate for a fully repaired historical supplement.

R3 – High; M:L194 and phase-table headings. Replace "primitive eigenvalue" by "eigenvalue on a primitive eigenvector"; define primitive and explicitly allow nonprimitive examples. Distinguish module rank and binary dimension. No new proof needed; existing proof and explicit counterexample settle the wording. Small effort.

R4 – Medium; M:L35–48, Lemma 4.1. Add a short explanation that records are bookkeeping, zero blocks are retained, the tree is finite, and complexity is query count rather than a uniform gate bound. Most of this is already present; consolidate rather than duplicate. No new mathematics. Small effort.

Stage 2: novelty and citation repairs

R5 – High; introduction and claim ledger. Write a source-by-source theorem comparison for adaptive BV and generalized DJ. Compare assumptions, output criterion, allowed histories, and conclusion, not just titles. Include S2 equation (47), S5 Lemma 4.1, S4 Section 4, and S10's different support-polynomial model. New literature research: yes. New mathematics: only if comparison exposes a real gap. Moderate effort.

R6 – Medium; bibliography. Update the confirmed published S1 record, add precise locators, verify remaining entries, and add the original problem sources where explanatory history is given. Do not fabricate publication metadata. Literature work: yes. Small effort.

R7 – High for a new-theorem submission; before submission. Obtain a genuinely independent specialist reading of Lemma 4.1/Theorem 4.2 and the closest prior theorem comparisons. This audit does not substitute for independent human peer review. It is a prudent project gate, not a claim that every journal formally requires a pre-submission human audit. Moderate effort.

Stage 3: structure and explanation

R8 – Medium; first page and Section 4. Center BV and add the $n = 2$ four-record example. Explain the difference from ordinary complex BV at the degree-one parity step. No new mathematics. Small effort.

R9 – Medium; Sections 5–7 and figures. Move ring and Simon material to appendix/supplement as appropriate; keep a short UNIQUE-SAT contrast; remove repetitive text boxes and internal publication status. No new mathematics. Small to moderate effort.

R10 – Optional; after Theorem 4.2. Consider the constructive identification-rank proposition after independent proof and priority review. It is not necessary for a correct preprint. New mathematical review and literature work: yes; fresh finite checks already supplied. Moderate effort.

Stage 4: submission polish

R11 – Medium; release package. Bundle only canonical sources, final PDF, bibliography, portable supplement, README, and manifest; keep historical reviews separate. Fix local-path dependencies. No new mathematics. Small to moderate effort.

R12 – Low; final PDF. Remove the forced page break where unnecessary, improve page-bottom transitions, check all table headings, and render again after edits. Build and visual QA required. Small effort.

Blockers and final publication gates

Fatal blockers

None identified in the stated numbered theorems. This is not a declaration that undiscovered errors or equivalent prior results cannot exist. An exact prior theorem could reduce the contribution to exposition even while all proofs remain correct.

Preprint blockers in the archive as received

The authorship-placeholder footnote must be resolved. The supplement and provenance statements must match what readers can actually access. Primitive-vector wording should be repaired at the same release pass. These are targeted repairs; a new general DJ optimum, an all-size Simon theorem, or the optional rank characterization is not required.

Submission blockers

The theorem-level contribution needs a concrete prior-art comparison rather than reliance on "no firstness claimed." Retained empirical claims need portable evidence and exact scopes. Administrative manuscript text and bibliographic incompleteness should be removed. The conditional novelty and significance judgment must be revisited if a matched theorem is found.

Nonblocking improvements

A worked $n = 2$ example, a shorter abstract, moving secondary material, a cleaner layout, the indicator-polynomial alternative proof, and the optional identification-rank proposition can improve the paper. They should not become excuses to broaden it indefinitely.

Final answers

Before public release: finalize the actual author metadata; correct primitive-vector terminology; either attach a self-contained supplement or remove/replace unsupported historical experimental and provenance claims; rebuild and inspect the resulting PDF. On that proof-centered route, the manuscript is ready after minor repairs.

Before peer review: additionally give precise theorem-level comparisons to the closest modal, polynomial, and nondeterministic-degree antecedents; justify the surviving contribution as a focused note; make the evidence package portable; and carry out a genuinely independent specialist read of the core proof and priority comparison. Submit the repaired paper, not a portfolio archive whose essential dependencies remain elsewhere.

Primary-source register

The locators below identify the material actually used. Bibliographic entries are not assertions that an entire long document was exhaustively audited.

S1. B. Schumacher and M. D. Westmoreland, *Modal Quantum Theory*, Foundations of Physics 42, 918–925 (2012). DOI: 10.1007/s10701-012-9650-z. Author preprint: arXiv:1010.2929. Publisher metadata and foundational model checked.

S2. B. Schumacher and M. D. Westmoreland, *Almost quantum theory* (2012), arXiv:1204.0701. Especially equation (47), PDF page 23, and Section 4.4, PDF pages 26–29: generalized modal operations and the common-kernel condition.

S3. R. P. James, G. Ortiz, and A. Sabry, *Quantum Computing over Finite Fields* (2011), arXiv:1101.3764. Reversible finite-field formulation and conclusion on PDF page 10; qualitative Deutsch antecedent.

S4. P. Diamond and B. Schumacher, *Cloning, deleting, and hiding in modal quantum theory* (2023), arXiv:2310.04397. Section 4, PDF pages 4–5: distinguishability and copies.

S5. R. Beals, H. Buhrman, R. Cleve, M. Mosca, and R. de Wolf, *Quantum Lower Bounds by Polynomials*, Journal of the ACM 48(4), 778–797 (2001). Author preprint, arXiv:quant-ph/9802049. Lemmas 4.1–4.2, PDF pages 4–5 of the preprint; amplitude versus probability degrees.

S6. A. J. Hanson, G. Ortiz, A. Sabry, and J. Willcock, *The Power of Discrete Quantum Theories* (2011), arXiv:1104.1630. PDF page 4, "Discrete Quantum Computing": odd-characteristic setting.

S7. Y.-T. Tai, *Discrete Quantum Theories and Computing*, Indiana University PhD dissertation (2019). Institutional full text. Sections 4.1 and 4.4, printed pages 30 and 42–45: complexified finite fields and discrete Deutsch. Parsed text was inspected; one screenshot request timed out and was not treated as visual evidence.

S8. N. de Beaudrap, *On computation with 'probabilities' modulo k* (2014), arXiv:1405.7381. State/valid-transformation definitions, including the unit-ideal formulation, and the uniform-circuit framework. This is not an exact oracle-bound citation.

S9. J. Willcock and A. Sabry, *Solving UNIQUE-SAT in a Modal Quantum Theory* (2011), arXiv:1102.3587. Section 3 and Figure 1 with the two correctness cases, PDF pages 3–5.

S10. R. de Wolf, *Nondeterministic Quantum Query and Communication Complexities*, SIAM Journal on Computing 32(3), 681–699 (2003), author version, arXiv:cs/0001014. Section 2.4, Theorem 2.3, printed pages 686–688: nondeterministic degree characterization in the ordinary quantum model.

S11. *Foundations of Physics*, current aims and scope, checked 27 September 2026. Used only for scope, not an acceptance forecast.